

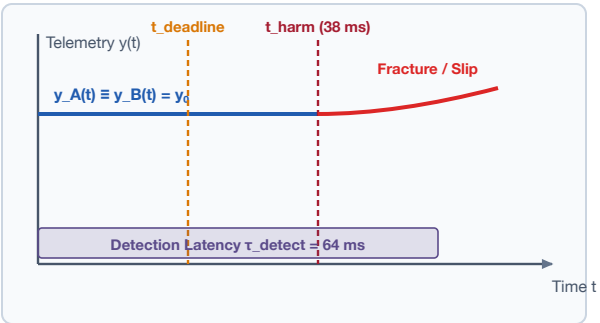
THE DUAL EPISTEMIC BOUNDARIES: DETECTOR DEADLINES & EXPOSURE WALLS

Why Empirical Testing Alone Cannot Settle Open-World Safety, Demanding Architectural Authority Limits and Closure Tests

1. OBSERVABILITY DEADLINE LIMIT

Identical Pre-Harm Telemetry

States S_A (Safe) vs S_B (Hazard) indistinguishable



- **State S_A vs S_B:** Nominal surface vs oil slick / internal void
- **Pre-Contact:** Zero optical or tactile surface signature
- **Late Detection:** Harm occurs before reflex engages ($38 \text{ ms} < 64 \text{ ms}$)
- **Physical Theorem:** Identical $y(t)$ implies safe intervention cannot
- **Epistemic Result:** exceed prior without structural force clamping

2. ASTRONOMICAL EXPOSURE WALL

$n \geq \ln(1/\alpha) / p$ at 95% Confidence

Required failure-free hours grows reciprocally with $1/p$

$p \leq 10^{-2}/\text{h}$ (Consumer App)

$n \geq 300 \text{ h}$

Time: 12.5 days

Tractable (Unit Test)

$p \leq 10^{-4}/\text{h}$ (Industrial Cobot)

$n \geq 3.0 \times 10^4 \text{ h}$

Time: 3.4 years

Affordable (Bench Rig)

$p \leq 10^{-6}/\text{h}$ (Autonomous Transit)

$n \geq 3.0 \times 10^6 \text{ h}$

Time: 342 years

Fleet Limit (Correlated)

$p \leq 10^{-9}/\text{h}$ (SIL-4 / ASIL-D)

$n \geq 3.0 \times 10^9 \text{ h}$

Time: 342,000 yrs

LOGICALLY INSUFFICIENT

WHY FLEET SAMPLING FAILS AT THE TAIL:

1. Correlated updates & identical software reset trial clock
2. Safety drivers censor crashes $\rightarrow$ missing counterfactuals
3. Simulators test only pre-programmed physics disturbances
4. Ultra-reliability demands structural containment!

3. RESIDUAL-CLAIMS REGISTER

Converting Caveats to Engineering Work

4 Mandatory Fields for Every Open Assumption

FIELD 1: UNSUPPORTED CLAIM

Brake distance $\leq 0.49 \text{ m}$ on concrete
Missing Observable: Surface friction $\mu < 0.15$

FIELD 2: OPERATIONAL CONTAINMENT

Structural velocity clamp $v \leq 0.9 \text{ m/s}$
Kinetic energy E_k reduced 83% ($1089 \text{ J} \rightarrow 182 \text{ J}$)

FIELD 3: ACCOUNTABLE OWNER

Lead Drivetrain Engineer owns parameter lock;
Accepts operational throughput penalty

FIELD 4: FALSIFIABLE CLOSURE TEST

Integrate polarimetric sensor + 1 kHz slip reflex;
Validated across $N=1000$ wet floor trials

FALSIFICATION & VERDICT GATE

Relocated Assumption? $\rightarrow$ REJECT / STOP

Independent Evidence? $\rightarrow$ RETIRE GAP & LIFT CLAMP